Tanzeel Rahman

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Education

Bachelor of Science in **Computer Science** — **University of South Florida** — Expected May **2027**
Judy Genshaft Honors College — Flexible to Dec 2027 | GPA: 3.8/4.0

Professional Experience

Technology Summer Analyst — **Citi** — May 2026 – Aug 2026

- Built an internal engineering analytics dashboard (**React**, **TypeScript**, **FastAPI**) for a **22**-member Production Support team
- Integrated **6,600+** Jira/ServiceNow tickets into the dashboard's reporting and analytics layer
- Reduced roughly an **hour** of status reporting per daily standup via self-service, drill-down visibility into tickets and incidents
- Translated stakeholder requirements into a production-ready analytics solution, iterating directly with engineering leadership through development, presentation, and adoption
- Used **Devin** and Citi's AI assistant Stylus (Claude Opus, Gemini) throughout the build, catching Devin hallucinating data past its pull limit and splitting large tasks into small, explicitly scoped pieces verified by hand
- Delivered the dashboard end-to-end and deployed it on **OpenShift** (**Kubernetes**), writing YAML deployment manifests and scaling, monitoring, and debugging pods

Software Engineer, Mobile — **University of South Florida** — Jan 2025 – Present

- Architected **C#** services for GPS tracking, speed computation, and route data collection using **Unity** and **Android SDK**
- Engineered an Android app for driving-behavior research, enabling authentication and **GPS** collection across **50** participants
- Implemented asynchronous processing, lifecycle-aware state management, and robust error handling for reliable field data collection
- Designed **Supabase**-backed authentication and data persistence workflows to manage participant records and research data

Software Systems Engineer Intern — **Moffitt Cancer Center** — May 2025 – Present

- Automated extraction and normalization of genomic reports using **Python** (Regex, **Pandas**), cutting processing time by **60%**
- Designed data pipelines standardizing mutated gene, TMB, and MSI fields across heterogeneous schemas for **10,000+** records
- Built reusable **Python** and **R** scripts for validation, analysis, and visualization across heterogeneous genomic datasets
- Collaborated directly with biomedical researchers to translate domain-specific scientific constraints into ML-ready datasets, producing analytic summaries for internal review and **NIH**-facing documentation

Data Engineering Intern — **Purdue University** — Jan 2025 – Dec 2025

- Engineered **Python**, **SQL**, and **Pandas** pipelines to clean, transform, and analyze AI-in-education research data
- Applied the Self-Regulated Learning (**SRL**) framework to identify behavioral trends across qualitative and quantitative datasets
- First-authored peer-reviewed research on generative AI in engineering education, published and presented at the **2026 ASEE Annual Conference & Exposition**

Backend Software Engineer Intern — **Telecommunications Ltd.** — May 2024 – Aug 2024

- Supported **500+** employees by building and maintaining high-availability **Node.js** and **Express.js** backend services
- Accelerated deployment cycles by **40%** by containerizing services with **Docker** and CI/CD pipelines using **GitHub Actions**
- Reduced API endpoint latency by **35%** by optimizing **SQL** queries and refactoring **Express.js** logic for internal dashboards
- Built **RESTful APIs** and **PostgreSQL** schemas (**Node.js**, **Express.js**) for real-time fault detection across **100K+** network logs

Projects

AI Workflow Orchestration & Incident Intelligence Platform | Python, FastAPI, PostgreSQL, pgvector, React, Docker

- Built a **DAG-based AI workflow engine** that orchestrates LLM calls, vector retrieval, conditional logic, HTTP requests, and Python functions across multi-step workflows
- Developed an **AI incident analysis system** that retrieves similar incidents and runbooks via **pgvector**, producing evidence-backed root-cause hypotheses
- Engineered a **PostgreSQL-backed job queue** for asynchronous workflow execution, coordinating concurrent workers with persistent state and per-node retries

Music Therapy App for Dementia Care | Kotlin, Jetpack Compose, Firebase, Firestore, Media3 (ExoPlayer), MVVM

- Engineered a native Android application for music therapy research (participant enrollment, session management, structured playback) for an elderly-patient study of **140** patients
- Implemented **Media3**-based playback, **Firebase** Authentication, and **Jetpack Compose** components, including emotion capture and progress tracking
- Designed a **Firestore** data model syncing music playback events with EmotiBit physiological data for **Python** analysis

Technical Skills

- **Languages**: Python, Java, TypeScript, JavaScript, SQL, C++, C, C#, Kotlin, R
- **Frameworks & Libraries**: React.js, FastAPI, Node.js, Express.js, Jetpack Compose, Flutter, Pandas
- **Databases & Cloud**: PostgreSQL, SQLite, pgvector, Supabase, Docker, Kubernetes, OpenShift, AWS
- **Developer Tools**: Linux, Git, GitHub Actions, ServiceNow, Jira, Tableau, Devin, Stylus
- **Software Engineering:** REST APIs, LLM Integration, Data Pipelines, CI/CD, Agile, Automated Testing